CS2201 Computer Organization & Architecture

L27: Address Mapping in DRAM



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Professor

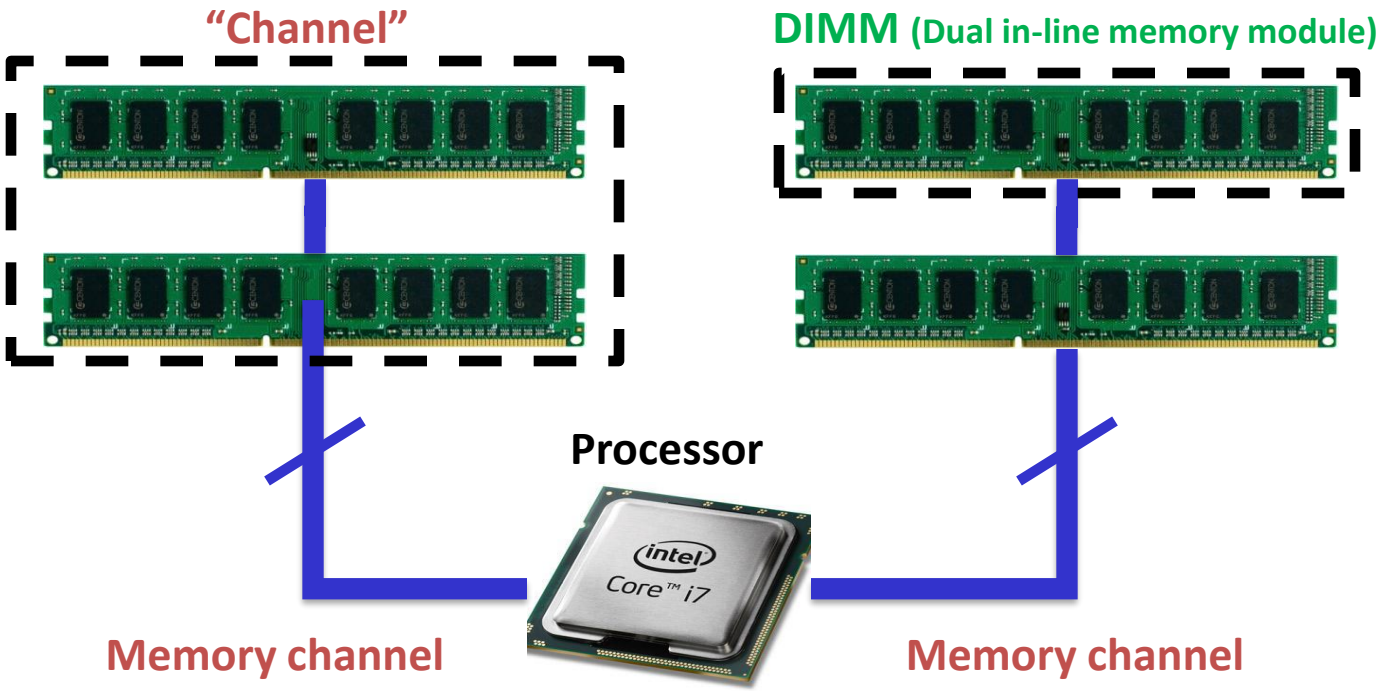
Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

DRAM Subsystem Organization

- Channel
- DIMM
- Rank
- Chip
- Bank
- Row
- Column
- B-Cell

The DRAM subsystem

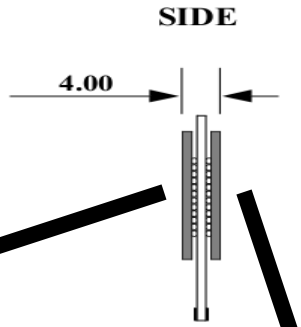


Breaking down a DIMM

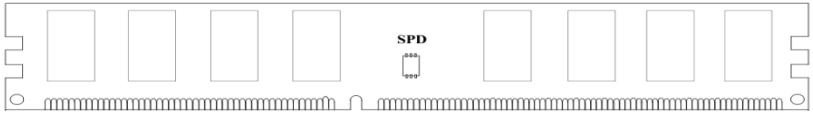
DIMM (Dual in-line memory module)



Side view



Front of DIMM



Back of DIMM

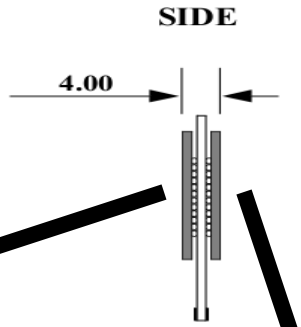


Breaking down a DIMM

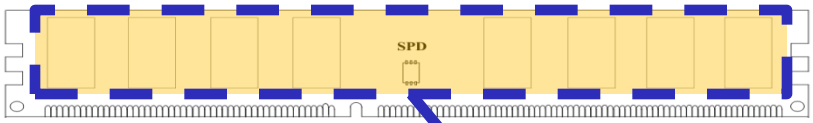
DIMM (Dual in-line memory module)



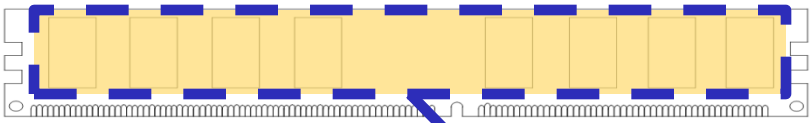
Side view



Front of DIMM



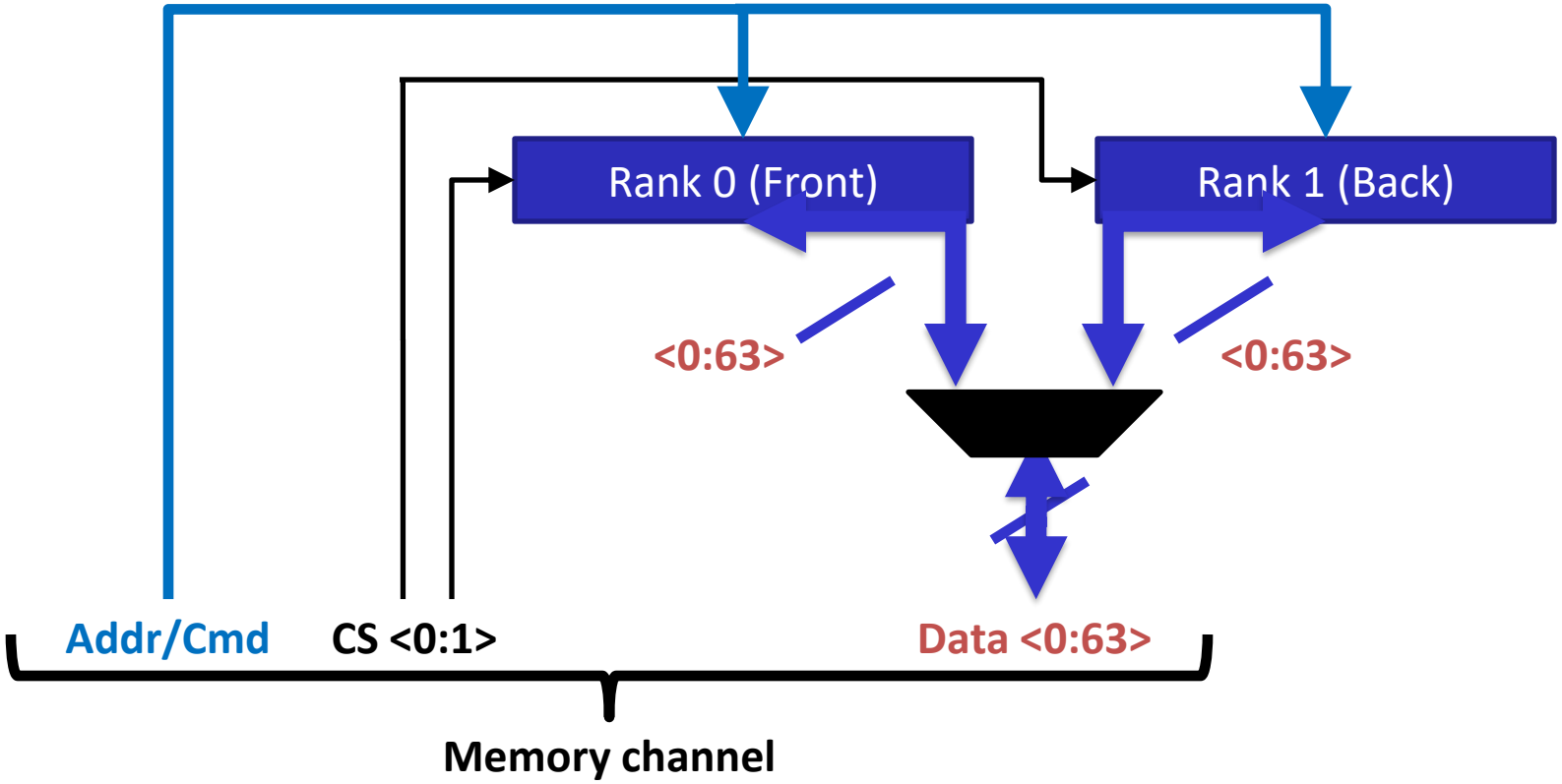
Back of DIMM



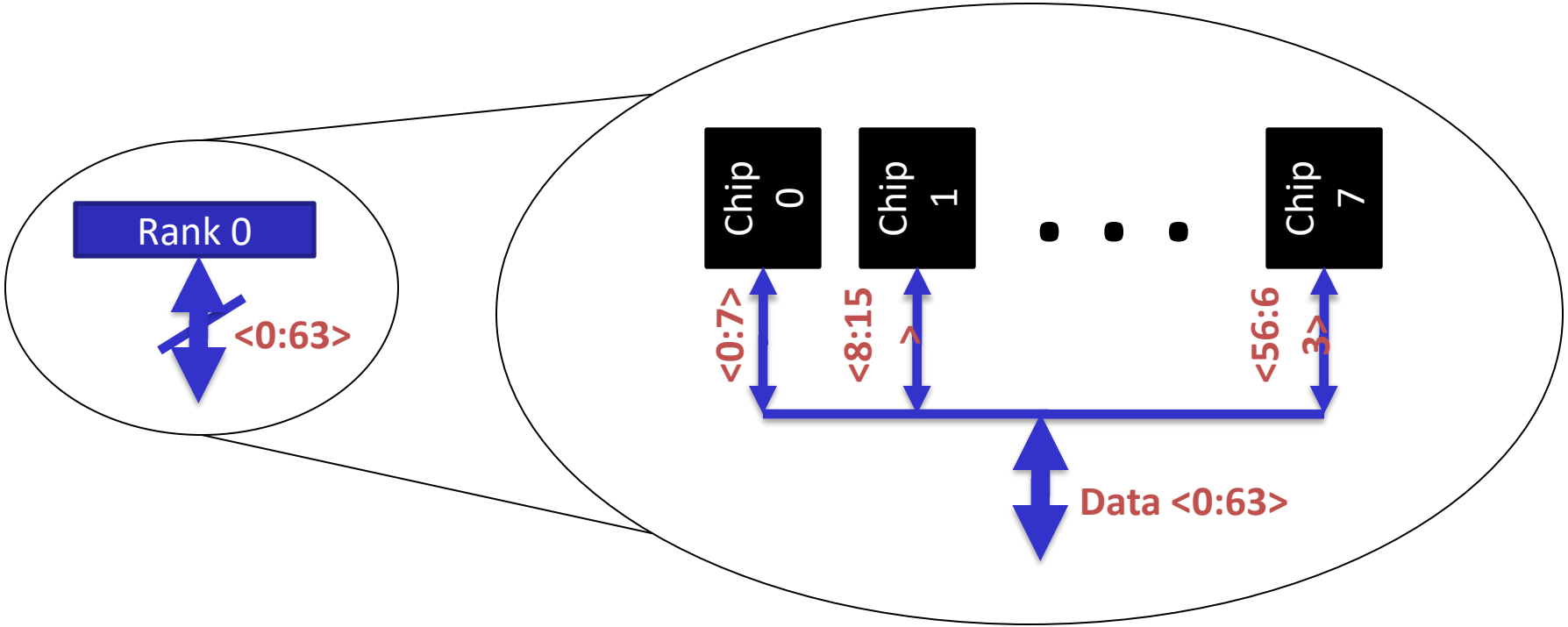
Rank 0: collection of 8 chips

Rank 1

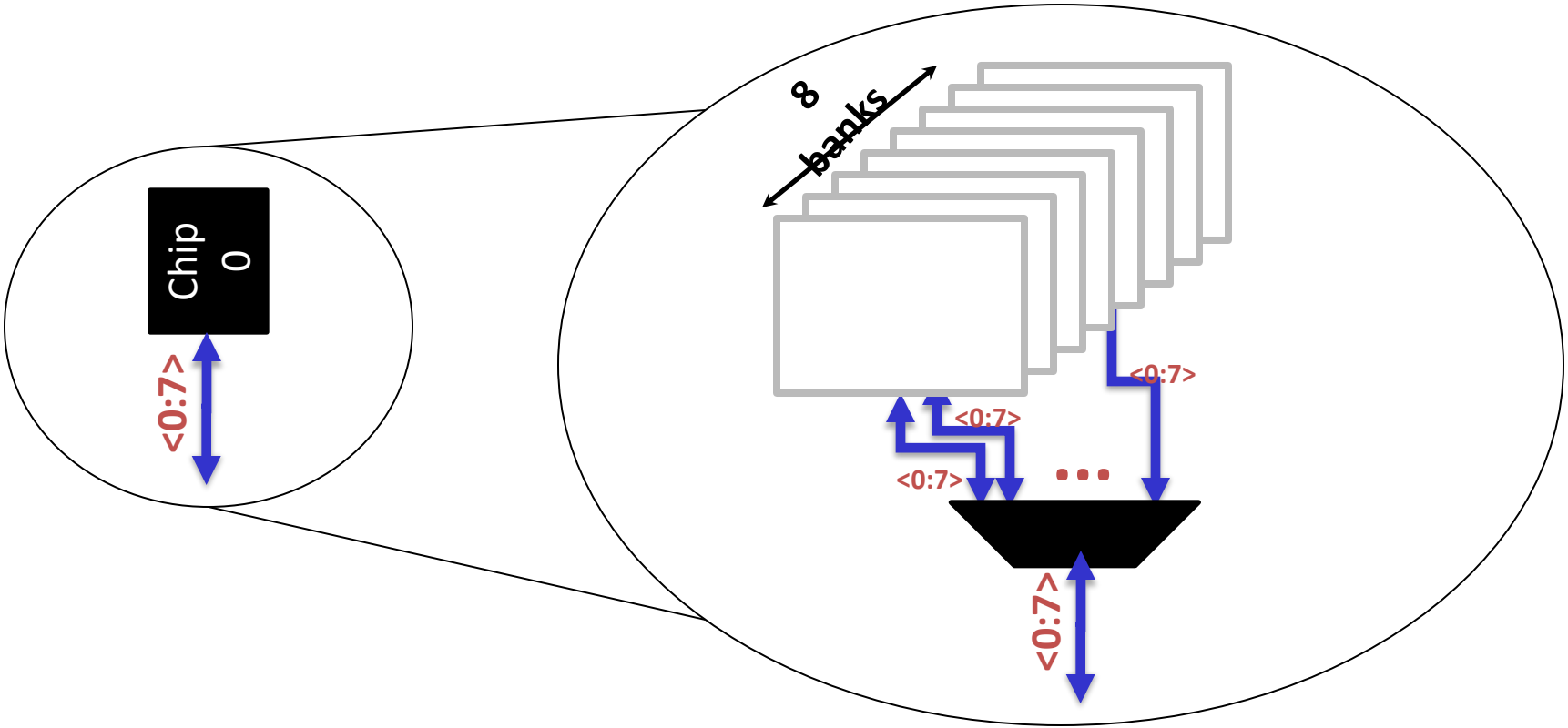
Rank


Rank 0 (Front)
Rank 1 (Back)
<0:63>
<0:63>
Addr/Cmd
CS <0:1>
Data <0:63>
Memory channel

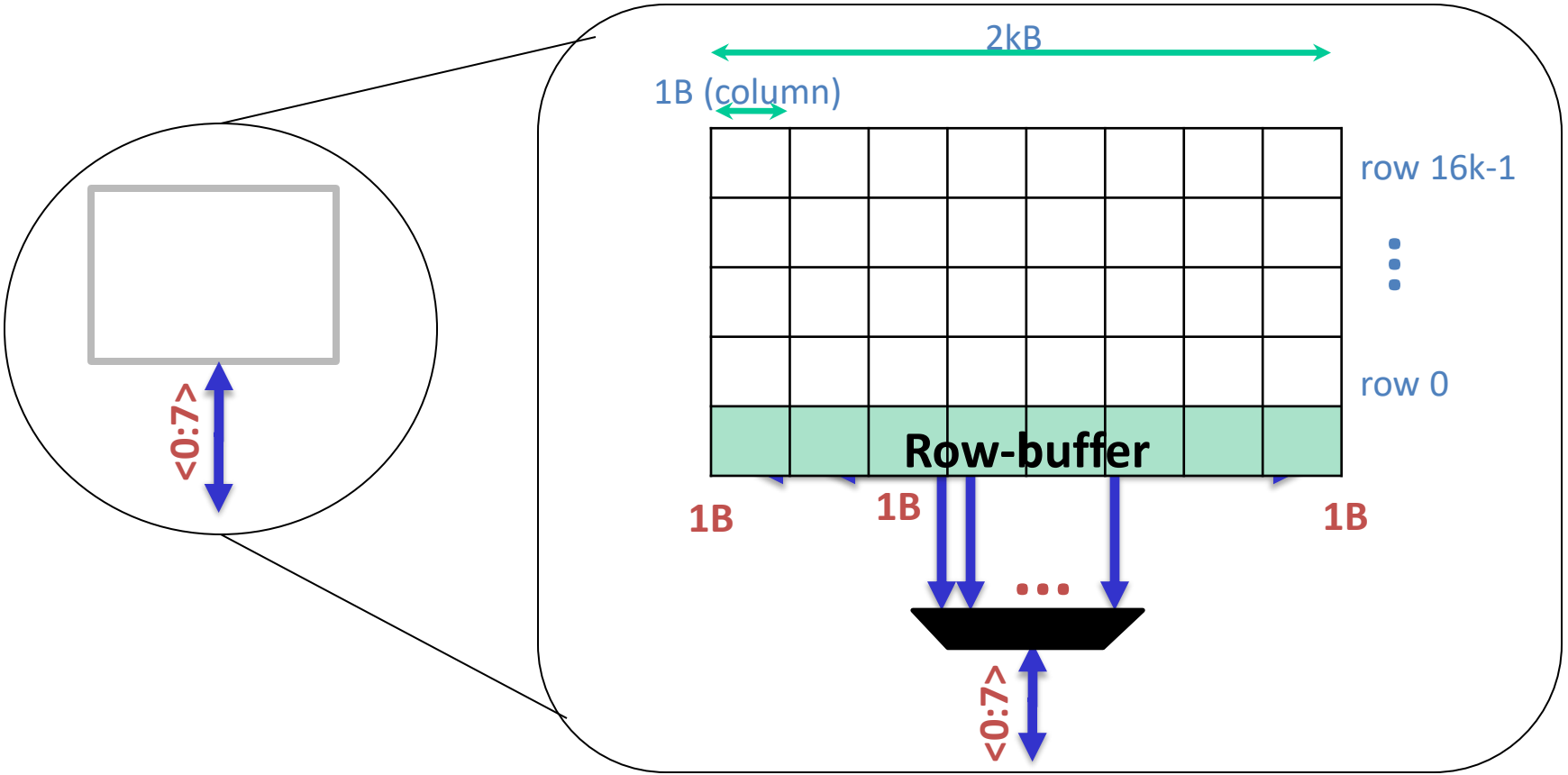
Breaking down a Rank



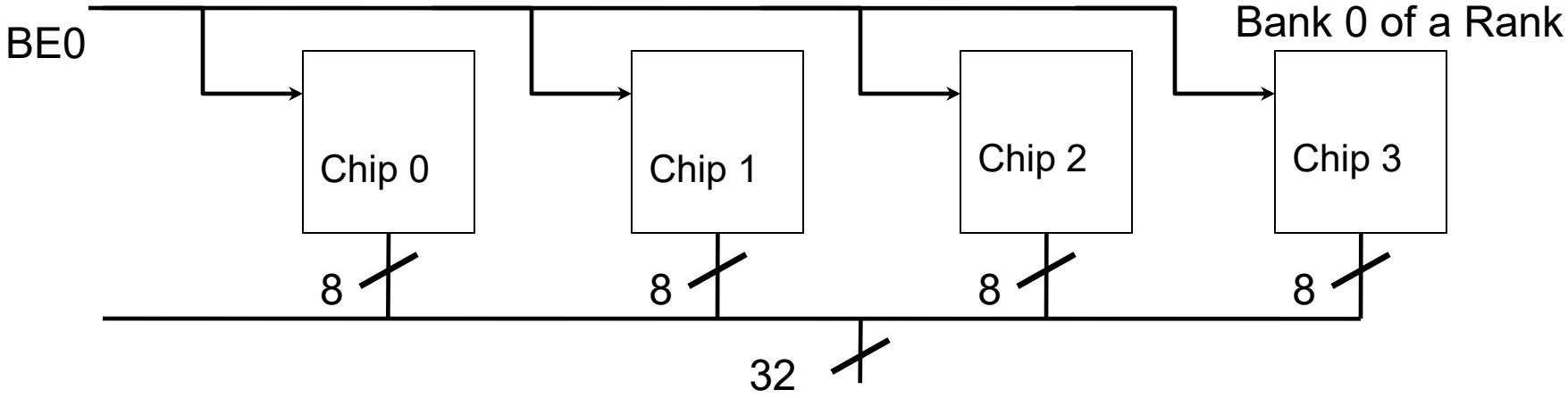
Breaking down a Chip



Breaking down a Bank



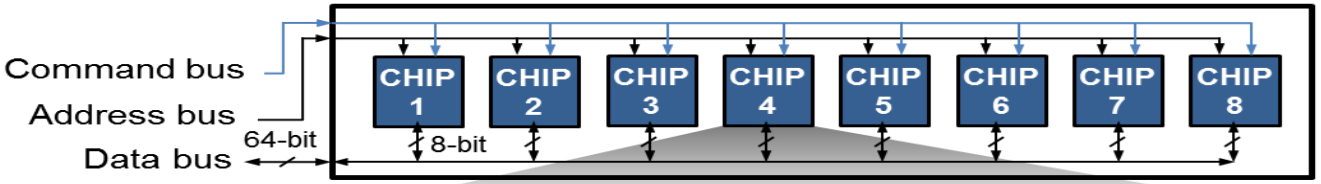
DRAM Rank



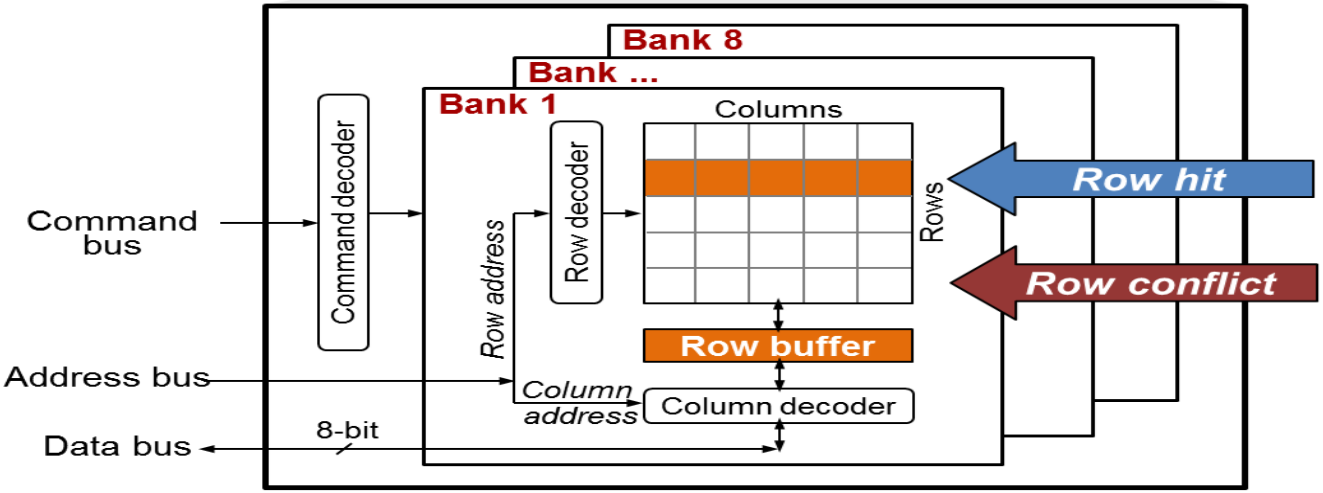
- Rank : A set of chips that respond to same command and same address at the same time but with different pieces of the requested data.
- Easy to produce 8 bit chip than 32 bit chip.
- Produce an 8 bit chip but control and operate them as a rank to get a 32 bit data in a single read.

DRAM Rank

DRAM Rank



DRAM Chip

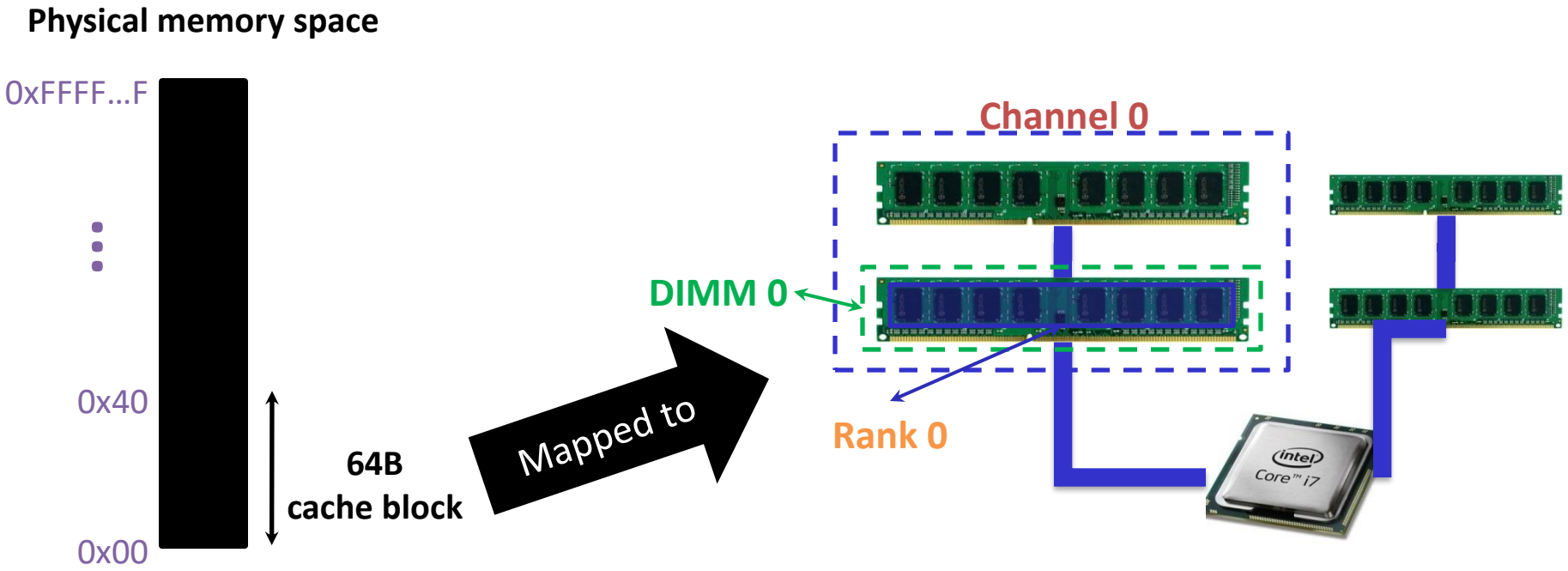


DRAM access latency varies depending on which row is stored in the row buffer

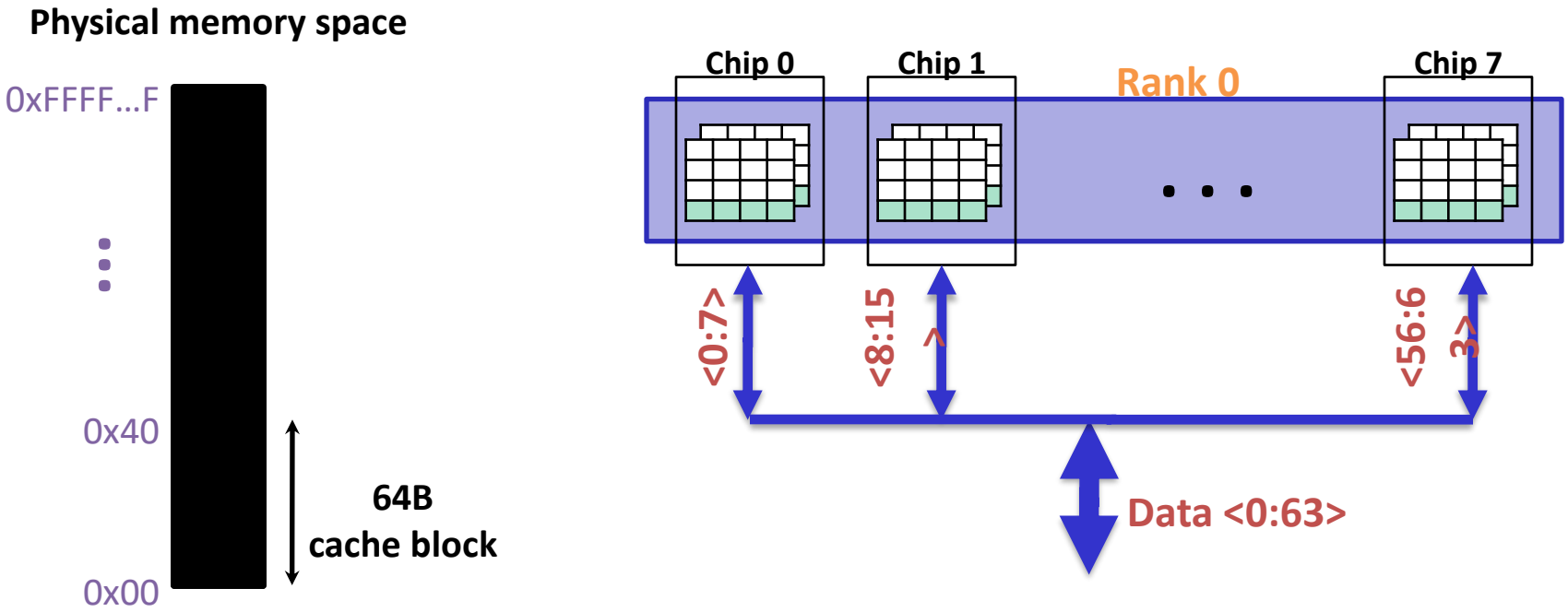
Basic DRAM Operation

- CPU → controller transfer time
- Controller latency
  - Queuing & scheduling delay at the controller
  - Access converted to basic commands
- Controller → DRAM transfer time : Bus latency (BL)
- DRAM bank latency
  - Simple CAS (column address strobe) if row is “open”
  - RAS (row address strobe) + CAS if array precharged
  - PRE + RAS + CAS (worst case)
- DRAM → Controller transfer time : Bus latency (BL)
- Controller to CPU transfer time

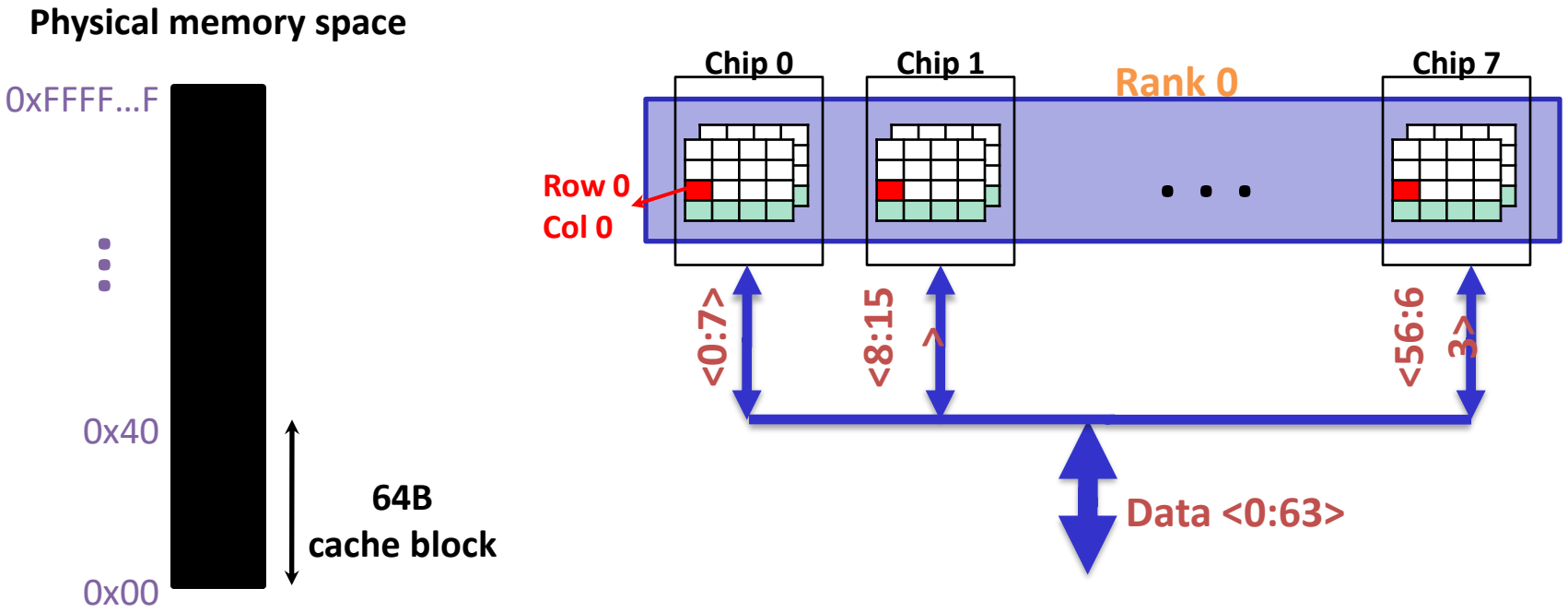
Transferring a cache block



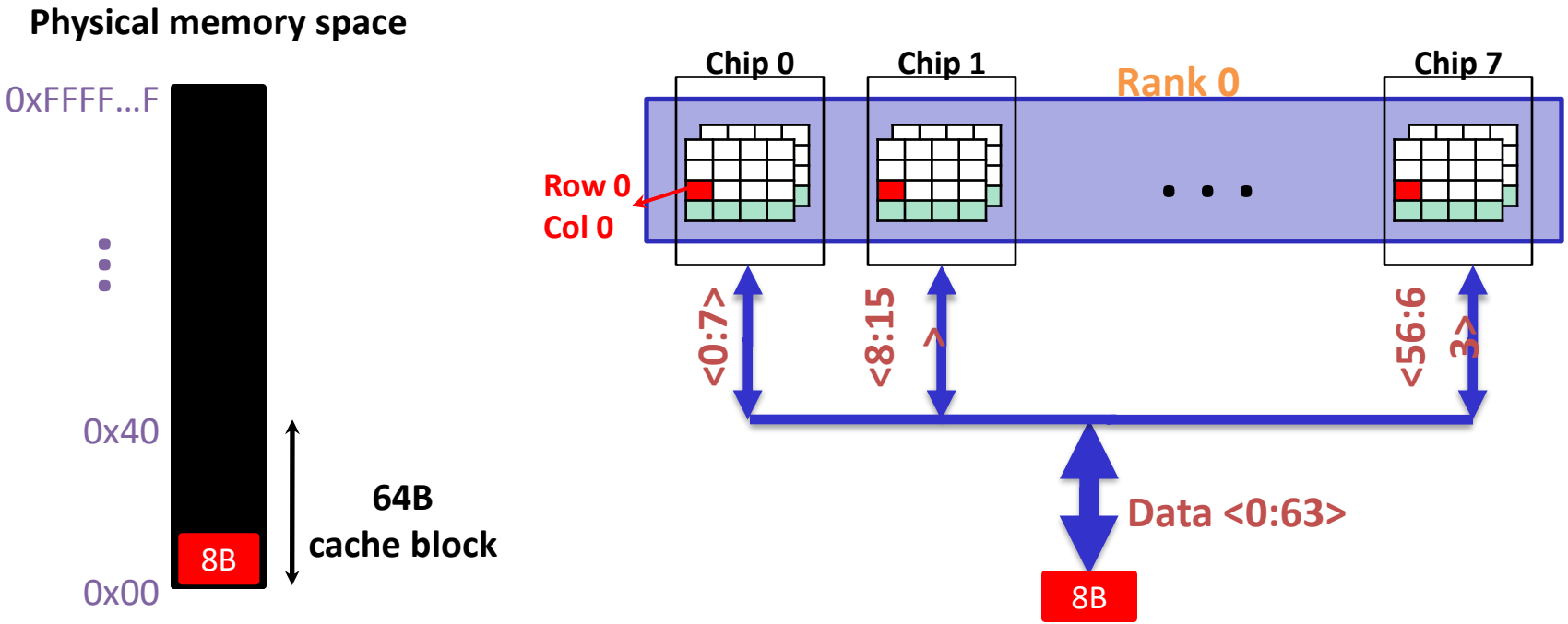
Transferring a cache block



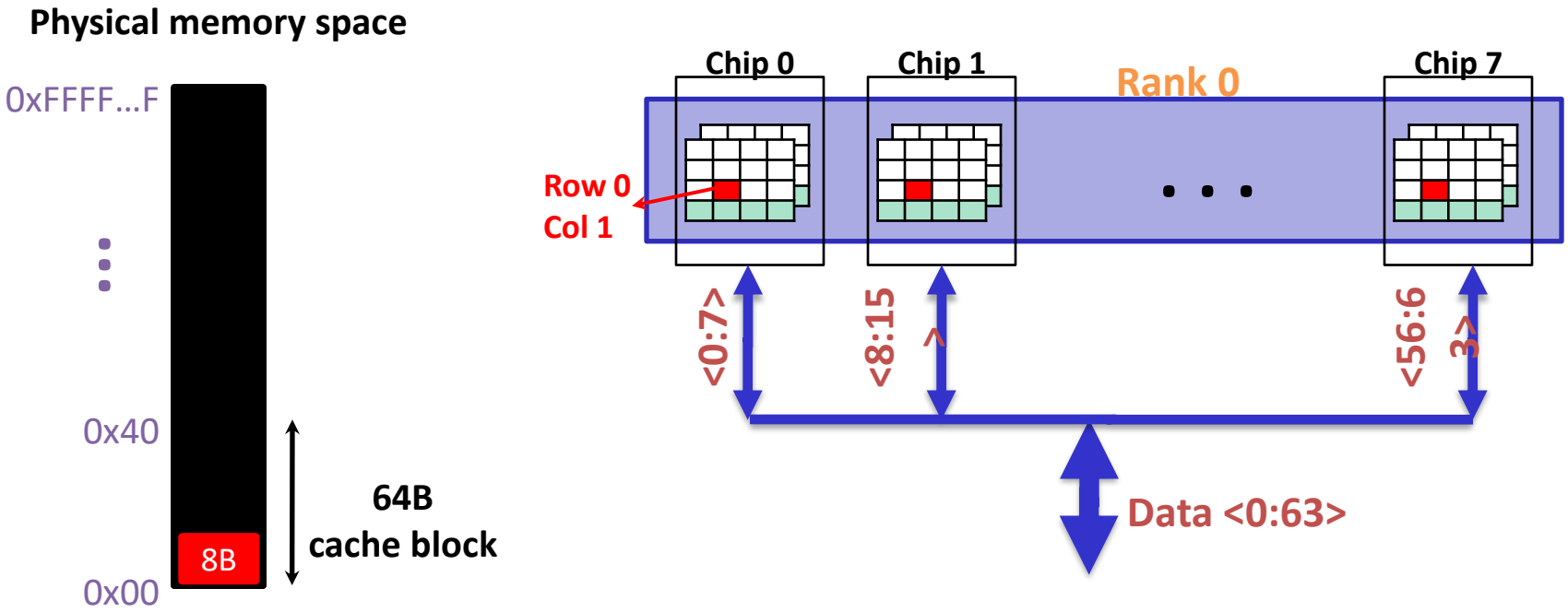
Transferring a cache block



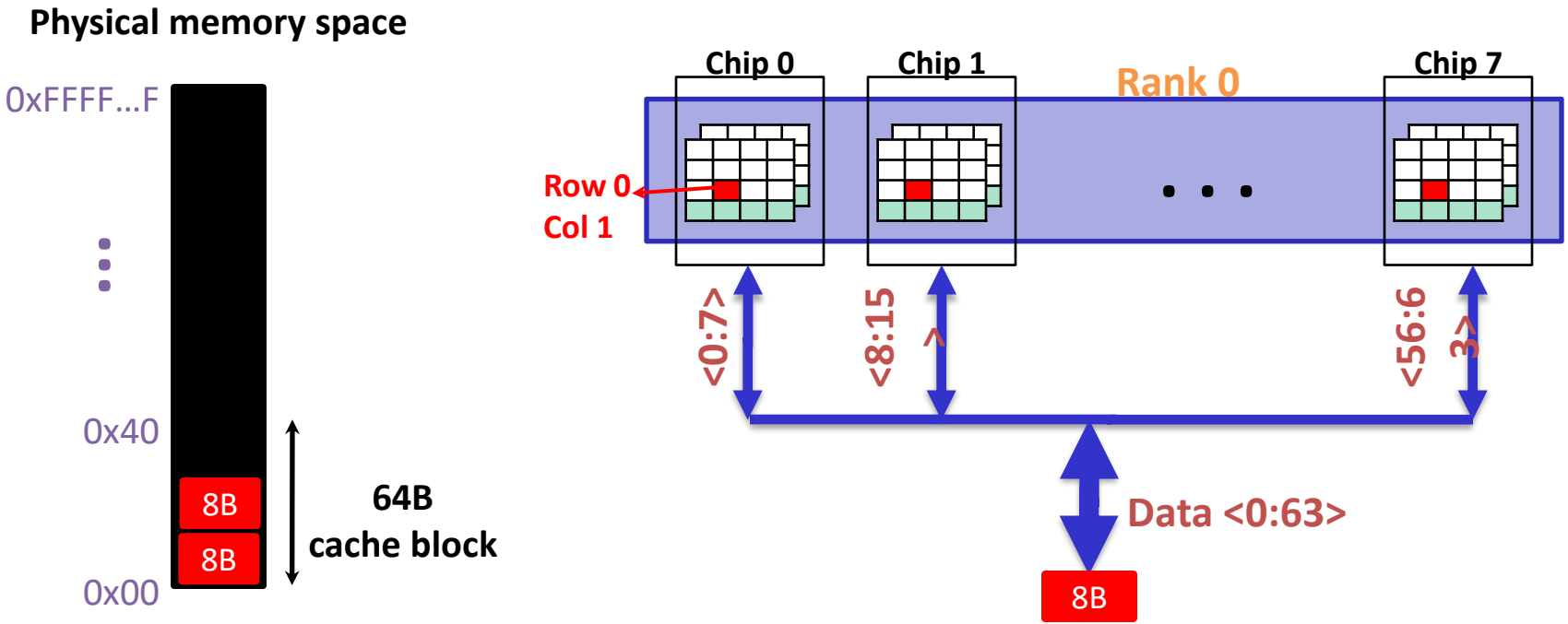
Transferring a cache block



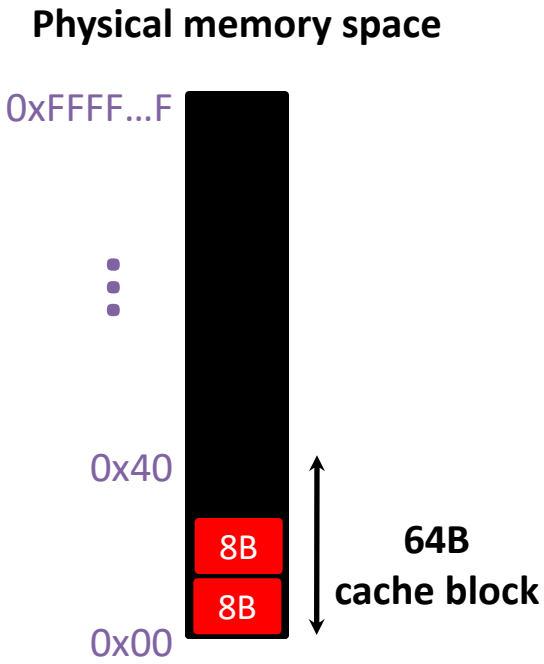
Transferring a cache block

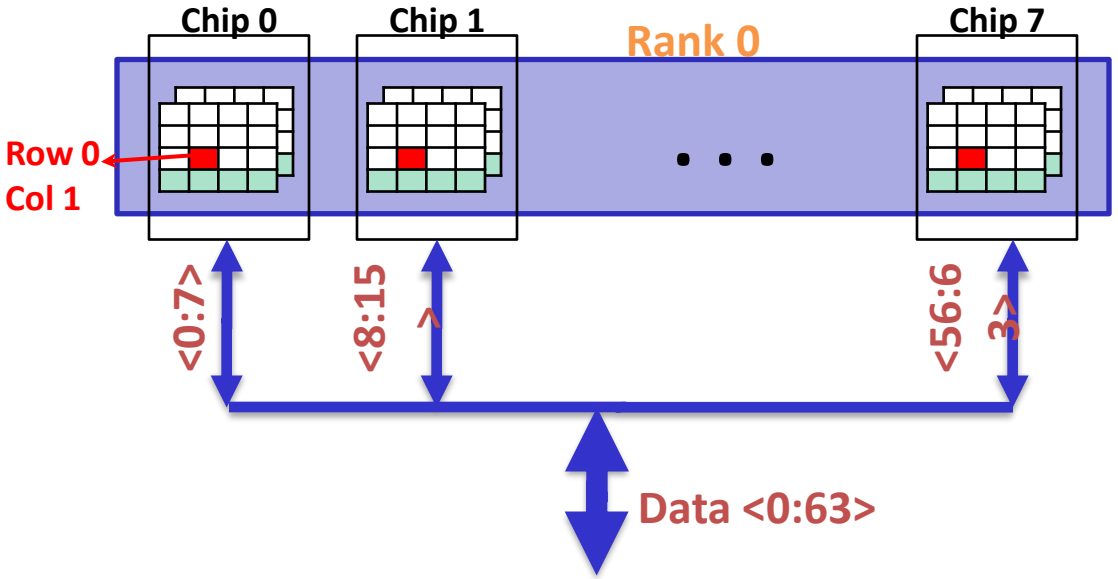


Transferring a cache block



Transferring a cache block





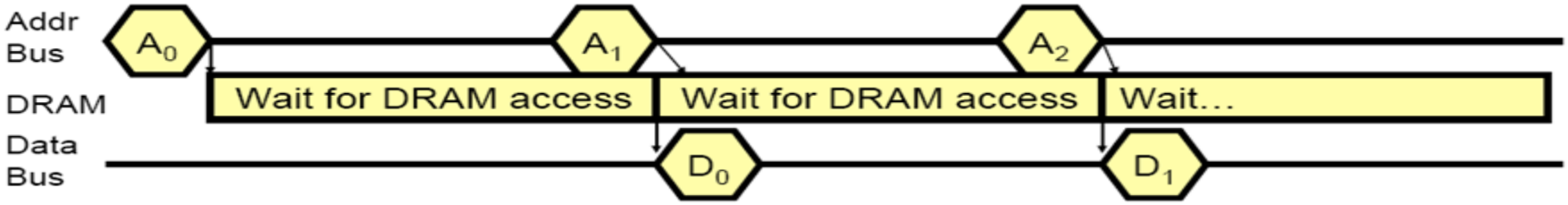
A 64B cache block takes 8 I/O cycles to transfer.

During the process, 8 columns are read sequentially.

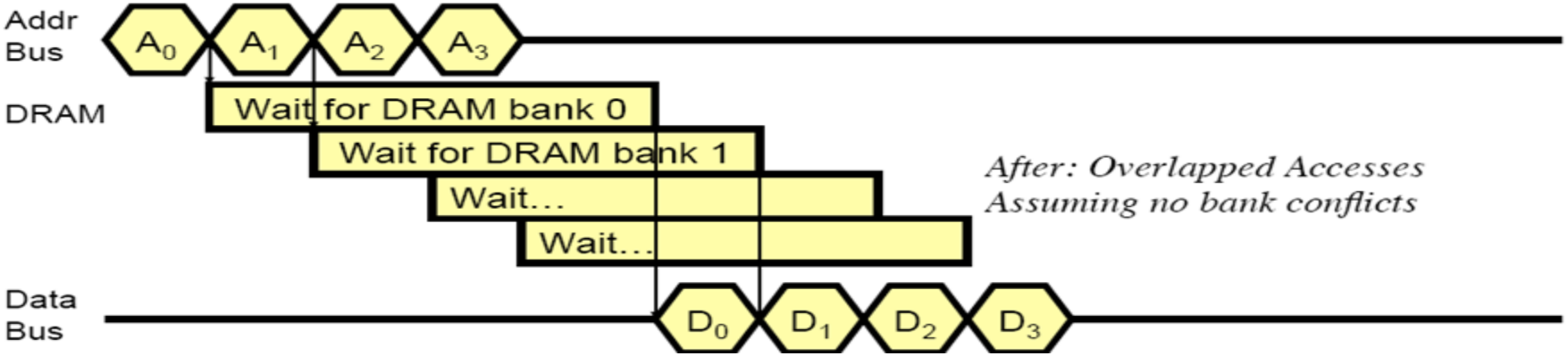
Multiple Banks and Channels

- **Multiple banks**
  - Enable concurrent DRAM accesses
  - Bits in address determine which bank an address resides in
- **Multiple independent channels**
  - Fully parallel as they have separate data buses
  - Increased bus bandwidth
  - More wires, area and power consumptions
  - More pins for on-chip memory controller
- Enabling more concurrency requires reducing
  - Bank conflicts, Channel conflicts

Multiple banks to reduce delay



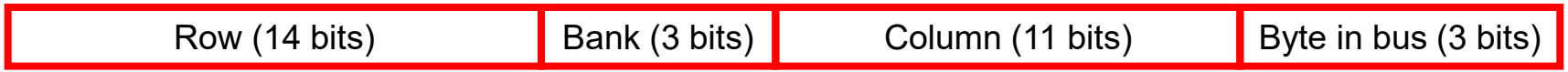
Before: No Overlapping
Assuming accesses to different DRAM rows



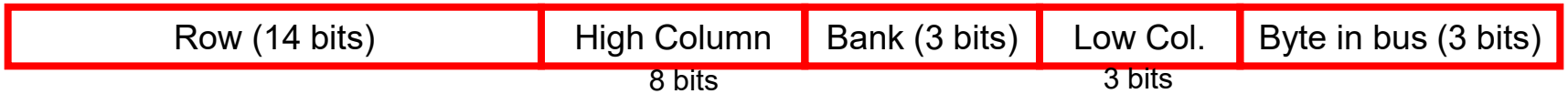
After: Overlapped Accesses
Assuming no bank conflicts

Address Mapping (Single Channel)

- Single-channel system, 8B memory bus
  - 2GB memory, 8 banks, 16K rows & 2K columns per bank
- Row interleaving
  - Consecutive rows of memory in consecutive banks
  - Accesses to consecutive cache blocks serviced in a pipelined manner



- Cache block interleaving
  - Consecutive cache block addresses in consecutive banks
  - 64 B cache blocks, Accesses to consecutive cache blocks in parallel



Address Mapping (Multiple Channels)

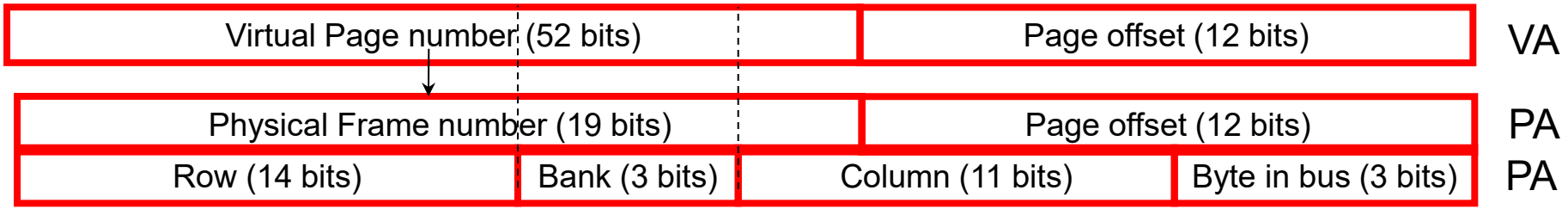
C	Row (14 bits)	Bank (3 bits)	Column (11 bits)	Byte in bus (3 bits)
Row (14 bits)	C	Bank (3 bits)	Column (11 bits)	Byte in bus (3 bits)
Row (14 bits)	Bank (3 bits)	C	Column (11 bits)	Byte in bus (3 bits)
Row (14 bits)	Bank (3 bits)	Column (11 bits)	C	Byte in bus (3 bits)

- Where are consecutive cache blocks?

C	Row (14 bits)	High Column 8 bits	Bank (3 bits)	Low Col. 3 bits	Byte in bus (3 bits)
Row (14 bits)	C	High Column 8 bits	Bank (3 bits)	Low Col. 3 bits	Byte in bus (3 bits)
Row (14 bits)	High Column 8 bits	C	Bank (3 bits)	Low Col. 3 bits	Byte in bus (3 bits)
Row (14 bits)	High Column 8 bits	Bank (3 bits)	C	Low Col. 3 bits	Byte in bus (3 bits)
Row (14 bits)	High Column 8 bits	Bank (3 bits)	Low Col. 3 bits	C	Byte in bus (3 bits)

Virtual To Physical Address Mapping

- Operating System influences where an address maps to in DRAM



- Operating system can influence which bank/channel/rank a virtual page is mapped to.
- It can perform page coloring to
  - Minimize bank conflicts
  - Minimize inter-application interference



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